

Lab Experiment 8: Conduct Z test for the Data Given using Python Objective:

AIM:

To test whether the average weight of a species of birds differs from 150 grams.

DESCRIPTION:

1. Null Hypothesis (H_0): The average weight of the birds is 150 grams.
2. Alternative Hypothesis (H_1): The average weight of the birds is not 150 grams.
3. Sample: Measure the weights of 30 birds randomly selected from the population.
4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.
5. Decision Rule: Use a significance level of $\alpha = 0.05$.

CODE:

```
import numpy as np

import scipy.stats as stats

sample_data = np.array([
    145, 148, 150, 147, 149, 151, 146, 152,
    150, 148, 149, 147, 153, 150, 151, 146,
    149, 152, 148, 150, 147, 151, 149, 150,
    146, 153, 148, 151, 150, 149
])

population_mean = 150

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print("Sample Mean:", sample_mean)

print("Sample Standard Deviation:", sample_std)

print("Z-statistic:", z_statistic)

print("P-value:", p_value)
```

```
if p_value < 0.05:
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```
    print("Reject the null hypothesis")
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else:
```

```
    print("Fail to reject the null hypothesis")
```

OUTPUT:

Sample Mean: 149.16666666666666

Sample Standard Deviation: 2.118609389936341

Z-statistic: -2.154410656139678

P-value: 0.031207981635716475

Reject the null hypothesis

RESULT:

There is no significant difference between the average bird weight and 150 grams at the 5% significance level